

ER Membrane Protein Classification (Types I–IV)

A Comparative Guide for CSIR NET Life Sciences
(Reference: Lodish & Cooper)

Expert High-Yield Reference Summary

Compiled for CSIR NET Life Sciences Preparation

1 Introduction and Theoretical Framework

Integral membrane proteins of the Endoplasmic Reticulum (ER) are classified according to:

- Orientation of the N- and C-termini
- Presence or absence of cleavable signal peptides
- Nature of topogenic sequences
- Mechanism of insertion through the **Sec61 translocon**

For the CSIR NET Life Sciences examination, understanding the following concepts is essential:

- **Positive-Inside Rule**
- **Signal-Anchor (SA) sequences**
- **Stop-Transfer Anchor (STA) sequences**
- Membrane topology prediction

2 Comprehensive Comparative Table

3 Core High-Yield Principles for CSIR NET

3.1 1. Positive-Inside Rule

The orientation of membrane proteins is strongly influenced by the distribution of positively charged amino acids (**Arginine and Lysine**) surrounding the hydrophobic transmembrane segment.

- The side containing more positive charges remains in the **cytosol**
- This rule helps determine final membrane topology

Examples:

- **Type II proteins:** Positive charges occur **before** the SA sequence $\Rightarrow$ N-terminus remains cytosolic
- **Type III proteins:** Positive charges occur **after** the SA sequence $\Rightarrow$ C-terminus remains cytosolic

3.2 2. Topogenic Sequence Definitions

- **Signal-Anchor (SA) Sequence:**

An internal hydrophobic sequence that:

- Directs the ribosome to the ER membrane
- Interacts with SRP
- Remains embedded as a transmembrane helix
- Is **not cleaved**

Feature	Type I	Type II	Type III	Type IV (Multipass)
Pass Status	Single-pass	Single-pass	Single-pass	Multipass membrane protein
N-terminus Location	ER lumen	Cytosol	ER lumen	Variable (IV-A or IV-B)
C-terminus Location	Cytosol	ER lumen	Cytosol	Depends on number of TM segments
N-terminal Signal Sequence	Present and cleavable	Absent	Absent	Absent
Topogenic Sequences	Stop-Transfer Anchor (STA)	Signal-Anchor (SA)	Signal-Anchor (SA)	Alternating SA and STA
Mechanism of Insertion	Insertion initiated by N-terminal signal peptide; STA halts translocation and anchors protein	Internal SA sequence acts as targeting signal and membrane anchor	Internal SA sequence acts as targeting signal and membrane anchor	Sequential insertion using alternating SA and STA sequences
Positive-Inside Rule	Positive residues usually remain on cytosolic side	Positive charges located before SA sequence	Positive charges located after SA sequence	Charge distribution determines loop orientation
Classic Examples	• Glycophorin A • LDL receptor • Influenza HA	• Transferrin receptor • Galactosyltransferase • Neuraminidase	• Cytochrome P450	• IV-A: GLUT, Connexin • IV-B: GPCRs (Rhodopsin)

Table 1: Structural and mechanistic comparison of Type I–IV ER membrane proteins

• **Stop-Transfer Anchor (STA) Sequence:**

A hydrophobic sequence that:

- Stops translocation through the translocon
- Causes lateral release into the lipid bilayer
- Forms a membrane-spanning segment

4 Quick Exam-Oriented Summary

Protein Type	Key Sequence	Orientation
Type I	Signal peptide + STA	N lumen / C cytosol
Type II	Internal SA	N cytosol / C lumen
Type III	Internal SA	N lumen / C cytosol
Type IV	Multiple SA + STA	Multipass topology